Eigen Values and Vectors

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Question : What can we conclude if a matrix $\mathbf{A}$ is symmetric?

Proof :

Case 1 : A is real and symmetric

Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be a real symmetric matrix, i.e.,

$$\mathbf{A} = \mathbf{A}^\top \tag{1}$$

Suppose λ is an eigenvalue of $\mathbf{A}$ with eigenvector $\mathbf{v} \neq \mathbf{0}$:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \tag{2}$$

Taking the conjugate transpose of $\mathbf{v}$ and multiplying on the left:

$$\mathbf{v}^*\mathbf{A}\mathbf{v} = \lambda\mathbf{v}^*\mathbf{v} \tag{3}$$

Since $\mathbf{A}$ is real symmetric, $\mathbf{A} = \mathbf{A}^* = \mathbf{A}^\top$, therefore:

$$\mathbf{v}^*\mathbf{A}\mathbf{v} = (\mathbf{A}\mathbf{v})^*\mathbf{v} = (\lambda\mathbf{v})^*\mathbf{v} = \bar{\lambda}\,\mathbf{v}^*\mathbf{v} \tag{4}$$

Equating both expressions for $\mathbf{v}^*\mathbf{A}\mathbf{v}$:

$$\lambda\mathbf{v}^*\mathbf{v} = \bar{\lambda}\,\mathbf{v}^*\mathbf{v} \tag{5}$$

$$(\lambda - \bar{\lambda})\mathbf{v}^*\mathbf{v} = \mathbf{0} \tag{6}$$

Since $\mathbf{v} \neq \mathbf{0}$, $\mathbf{v}^*\mathbf{v} > 0$. Hence:

$$\lambda = \bar{\lambda} \tag{7}$$

Therefore, every eigenvalue λ of $\mathbf{A}$ is real.

Eigenvectors corresponding to distinct eigenvalues are orthogonal

Let

$$\mathbf{A}\mathbf{u} = \alpha\mathbf{u} \tag{8}$$

$$\mathbf{A}\mathbf{w} = \beta\mathbf{w} \tag{9}$$

Where $\alpha \neq \beta$ are distinct eigenvalues.Multiplying the first equation on the left by $\mathbf{w}^\top$:

$$\mathbf{w}^\top\mathbf{A}\mathbf{u} = \alpha\mathbf{w}^\top\mathbf{u} \tag{10}$$

Since $\mathbf{A}$ is symmetric, $\mathbf{A} = \mathbf{A}^\top$. Hence:

$$\mathbf{w}^\top\mathbf{A}\mathbf{u} = (\mathbf{A}\mathbf{w})^\top\mathbf{u} = (\beta\mathbf{w})^\top\mathbf{u} = \beta\mathbf{w}^\top\mathbf{u} \tag{11}$$

Equating both forms:

$$\alpha \mathbf{w}^\top \mathbf{u} = \beta \mathbf{w}^\top \mathbf{u} \tag{12}$$

Simplifying:

$$(\alpha - \beta)\mathbf{w}^\top \mathbf{u} = 0 \tag{13}$$

Since $\alpha \neq \beta$, it follows that:

$$\mathbf{w}^\top \mathbf{u} = 0 \tag{14}$$

Hence, eigenvectors corresponding to distinct eigenvalues are orthogonal.

Proving that Eigen Vector is real
Suppose $\lambda \in \mathbb{R}$ is an eigenvalue of $\mathbf{A}$ with a (possibly complex) eigenvector $\mathbf{v} \in \mathbb{C}^n$, satisfying

$$\mathbf{A}\mathbf{v} = \lambda \mathbf{v} \tag{15}$$

Considering the conjugate eigenvector
Since $\mathbf{A}$ is real, taking the complex conjugate of both sides gives

$$\mathbf{A}\bar{\mathbf{v}} = \bar{\lambda}\bar{\mathbf{v}} \tag{16}$$

But for a real symmetric matrix, $\lambda = \bar{\lambda}$ (as proven earlier). Hence,

$$\mathbf{A}\bar{\mathbf{v}} = \lambda\bar{\mathbf{v}} \tag{17}$$

Thus, $\bar{\mathbf{v}}$ is also an eigenvector corresponding to the same eigenvalue λ.

Constructing real vectors from v and $\bar{\mathbf{v}}$
Define

$$\mathbf{u} = \mathbf{v} + \bar{\mathbf{v}} \tag{18}$$

$$\mathbf{w} = i(\mathbf{v} - \bar{\mathbf{v}}) \tag{19}$$

Both $\mathbf{u}$ and $\mathbf{w}$ are **real vectors**, because $\mathbf{u}$ is the sum of a vector and its conjugate, and $\mathbf{w}$ is i times their difference.

Applying A to u and w
Since $\mathbf{A}$ is real and linear,

$$\mathbf{A}\mathbf{u} = \mathbf{A}(\mathbf{v} + \bar{\mathbf{v}}) = \mathbf{A}\mathbf{v} + \mathbf{A}\bar{\mathbf{v}} = \lambda\mathbf{v} + \lambda\bar{\mathbf{v}} = \lambda(\mathbf{v} + \bar{\mathbf{v}}) = \lambda\mathbf{u} \tag{20}$$

Similarly,

$$\mathbf{A}\mathbf{w} = \mathbf{A}\big(i(\mathbf{v} - \bar{\mathbf{v}})\big) = i(\mathbf{A}\mathbf{v} - \mathbf{A}\bar{\mathbf{v}}) = i(\lambda\mathbf{v} - \lambda\bar{\mathbf{v}}) = \lambda i(\mathbf{v} - \bar{\mathbf{v}}) = \lambda\mathbf{w} \tag{21}$$

Hence we get

$$\mathbf{A}\mathbf{u} = \lambda\mathbf{u} \tag{22}$$

$$\mathbf{A}\mathbf{w} = \lambda\mathbf{w} \tag{23}$$

Real eigenvectors
If $\mathbf{u} \neq \mathbf{0}$, then $\mathbf{u}$ is a real eigenvector corresponding to eigenvalue λ. If $\mathbf{w} \neq \mathbf{0}$, then $\mathbf{w}$ is a real eigenvector corresponding to eigenvalue λ.

Conclusion : For any real symmetric matrix $\mathbf{A}$,

1) All eigenvalues of $\mathbf{A}$ are **real**.
2) Eigenvectors corresponding to distinct eigenvalues are **orthogonal**.
3) For every eigenvalue λ of a real symmetric matrix $\mathbf{A}$, there exists a **real eigenvector**. Hence, the eigenvectors of real symmetric matrices are real.

Case 2 : A is complex and symmetric

Let $\mathbf{A} \in \mathbb{C}^{n\times n}$ be a complex symmetric matrix satisfying

$$\mathbf{A} = \mathbf{A}^{\top} \tag{24}$$

Note that $\mathbf{A}$ is not necessarily Hermitian since in general $\mathbf{A} \neq \mathbf{A}^{*}$.

$$\mathbf{A} \neq \mathbf{A}^{*} \tag{25}$$

Eigenvalues of A may be complex
Let $\lambda \in \mathbb{C}$ be an eigenvalue of $\mathbf{A}$ with eigenvector $\mathbf{v} \neq \mathbf{0}$ such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \tag{26}$$

Define scalar z as :

$$z = \mathbf{v}^{*}\mathbf{A}\mathbf{v} \tag{27}$$

Taking complex conjugate transpose on both sides ,

$$z^{*} = \mathbf{v}^{*}\mathbf{A}^{*}\mathbf{v} \tag{28}$$

If $\mathbf{A} = \mathbf{A}^{*}$, then

$$z = z^{*} \tag{29}$$

$$z = \bar{z} \text{ as z is a scalar} \tag{30}$$

Therefore z is a real scalar value

For Hermitian matrices ($\mathbf{A} = \mathbf{A}^{*}$), the quantity $\mathbf{v}^{*}\mathbf{A}\mathbf{v}$ is always real. However, for complex symmetric matrices, $\mathbf{A} \neq \mathbf{A}^{*}$, so

$$\mathbf{v}^{*}\mathbf{A}\mathbf{v} \text{ may not be real.} \tag{31}$$

The Rayleigh quotient is

$$\frac{\mathbf{v}^{*}\mathbf{A}\mathbf{v}}{\mathbf{v}^{*}\mathbf{v}} = \lambda \tag{32}$$

Since $\mathbf{v}^{*}\mathbf{A}\mathbf{v}$ may be complex, λ may also be complex.
Eigenvectors corresponding to distinct eigenvalues need not be orthogonal
For complex symmetric matrices, we have only $\mathbf{A} = \mathbf{A}^{\top}$, so the Hermitian symme try property does not hold.
Thus, for two eigenpairs $(\lambda_1, \mathbf{v}_1)$ and $(\lambda_2, \mathbf{v}_2)$ with $\lambda_1 \neq \lambda_2$, we **cannot guarantee**

$$\mathbf{v}_1^{*}\mathbf{v}_2 = 0 \tag{33}$$

Hence, eigenvectors corresponding to distinct eigenvalues **need not be orthogonal and are not guaranteed to be real**.
Conclusion : For a complex symmetric matrix $\mathbf{A}$:

1) The eigenvalues **can be complex**.
2) The eigenvectors corresponding to distinct eigenvalues are **not necessarily orthogonal and need not be real**.

But if $\mathbf{A} = \mathbf{A}^{*}$ then we get the same result as **case 1**

Examples to Support the Proof
Example 1 : Real Symmetric Matrix
(**a**) Let

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \tag{34}$$

Clearly, $\mathbf{A} = \mathbf{A}^\top$, hence $\mathbf{A}$ is real symmetric.
Finding Eigenvalues
The characteristic equation is:

$$\left|\mathbf{A} - \lambda \mathbf{I}\right| = 0 \tag{35}$$

Substituting $\mathbf{A}$,

$$\left|\begin{pmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{pmatrix}\right| = 0 \tag{36}$$

$$\begin{pmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{pmatrix} \xleftrightarrow{R_2 \to R_2 + \frac{1}{2-\lambda}R_1} \begin{pmatrix} 2-\lambda & -1 & 0 \\ 0 & \frac{(2-\lambda)^2-1}{2-\lambda} & -1 \\ 0 & -1 & 2-\lambda \end{pmatrix} \xleftrightarrow{R_3 \to R_3 + \frac{(2-\lambda)}{(2-\lambda)^2-1}R_2} \begin{pmatrix} 2-\lambda & -1 & 0 \\ 0 & \frac{(2-\lambda)^2-1}{2-\lambda} & -1 \\ 0 & 0 & \frac{(2-\lambda)^3-4(2-\lambda)}{(2-\lambda)^2-1} \end{pmatrix} \tag{37}$$

$$(2-\lambda)\Big(\frac{(2-\lambda)^2-1}{2-\lambda}\Big)\Big(\frac{(2-\lambda)^3-4(2-\lambda)}{(2-\lambda)^2-1}\Big) = 0 \tag{38}$$

Simplifying,

$$(2-\lambda)((2-\lambda)^2-2) = 0 \tag{39}$$

Hence, the eigenvalues are:

$$\lambda_1 = 2, \;\; \lambda_2 = 2 + \sqrt{2}, \;\; \lambda_3 = 2 - \sqrt{2} \tag{40}$$

All eigenvalues are real, confirming positive definiteness.

Eigenvectors: Solving $(\mathbf{A} - \lambda_i \mathbf{I})\mathbf{v}_i = 0$ gives:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \;\; \mathbf{v}_2 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}, \;\; \mathbf{v}_3 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \tag{41}$$

Thus, eigenvectors are real and mutually orthogonal.

Example 2 : Complex Symmetric Matrix
(**a**) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1+i \\ 1+i & 1 \end{pmatrix} \tag{42}$$

Here, $\mathbf{A} = \mathbf{A}^\top$ but $\mathbf{A} \neq \mathbf{A}^*$, so $\mathbf{A}$ is complex symmetric but not Hermitian.
Finding Eigenvalues

$$\left|\mathbf{A} - \lambda\mathbf{I}\right| = 0 \tag{43}$$

$$\left|\begin{pmatrix} 1-\lambda & 1+i \\ 1+i & 1-\lambda \end{pmatrix}\right| = 0 \tag{44}$$

$$\begin{vmatrix} 1-\lambda & 1+i \\ 1+i & 1-\lambda \end{vmatrix} \xleftrightarrow{R_2 \to R_2 - \frac{1+i}{1-\lambda}R_1} \begin{vmatrix} 1-\lambda & 1+i \\ 0 & (1-\lambda) - \frac{(1+i)^2}{1-\lambda} \end{vmatrix} \tag{45}$$

$$(1-\lambda)^2 - (1+i)^2 = 0 \tag{46}$$

$$(1-\lambda)^2 - 2i = 0 \tag{47}$$

$$1 - \lambda = \pm\sqrt{2i} \quad \Rightarrow \quad \lambda = 1 \mp \sqrt{2i} \tag{48}$$

Hence eigenvalues are:

$$\lambda_1 = 1 - \sqrt{2i}, \quad \lambda_2 = 1 + \sqrt{2i} \tag{49}$$

Eigenvectors:
Eigenvalue $\lambda_1 = 1 - \sqrt{2i}$:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -\frac{\sqrt{2i}}{1+i} \end{pmatrix} \tag{50}$$

Eigenvalue $\lambda_2 = 1 + \sqrt{2i}$:

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ \frac{\sqrt{2i}}{1+i} \end{pmatrix} \tag{51}$$

Summary:

$$\lambda_1 = 1 - \sqrt{2i}, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ -\frac{\sqrt{2i}}{1+i} \end{pmatrix} \tag{52}$$

$$\lambda_2 = 1 + \sqrt{2i}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ \frac{\sqrt{2i}}{1+i} \end{pmatrix} \tag{53}$$

The Eigen vectors are **complex and not orthogonal** (**b**) Let

$$\mathbf{A} = \begin{pmatrix} 1 & i & 0 \\ i & 1 & i \\ 0 & i & 1 \end{pmatrix} \tag{54}$$

Here, $\mathbf{A} = \mathbf{A}^\top$ but $\mathbf{A} \neq \mathbf{A}^*$, so $\mathbf{A}$ is complex symmetric but not Hermitian.
Finding Eigenvalues

$$\left|\mathbf{A} - \lambda\mathbf{I}\right| = 0 \tag{55}$$

$$\left|\begin{pmatrix} 1-\lambda & i & 0 \\ i & 1-\lambda & i \\ 0 & i & 1-\lambda \end{pmatrix}\right| = 0 \tag{56}$$

$$\begin{pmatrix} 1-\lambda & i & 0 \\ i & 1-\lambda & i \\ 0 & i & 1-\lambda \end{pmatrix} \xleftrightarrow{R_2 \to R_2 - \frac{i}{1-\lambda}R_1} \begin{pmatrix} 1-\lambda & i & 0 \\ 0 & \frac{(1-\lambda)^2+1}{1-\lambda} & i \\ 0 & i & 1-\lambda \end{pmatrix} \xleftrightarrow{R_3 \to R_3 - \frac{i(1-\lambda)}{(1-\lambda)^2+1}R_2} \begin{pmatrix} 1-\lambda & i & 0 \\ 0 & \frac{(1-\lambda)^2+1}{1-\lambda} & i \\ 0 & 0 & \frac{(1-\lambda)^3+2(1-\lambda)}{(1-\lambda)^2+1} \end{pmatrix} \tag{57}$$

Now determinant = product of diagonal entries:

$$(1-\lambda)\Big(\frac{(1-\lambda)^2+1}{1-\lambda}\Big)\Big(\frac{(1-\lambda)^3+2(1-\lambda)}{(1-\lambda)^2+1}\Big) = 0 \tag{58}$$

Simplifying,

$$(1-\lambda)((1-\lambda)^2+2) = 0 \tag{59}$$

Hence eigenvalues are:

$$\lambda_1 = 1,\ \lambda_2 = 1 + i\sqrt{2},\ \lambda_3 = 1 - i\sqrt{2} \tag{60}$$

Conclusion : For a complex symmetric matrix,
1) Eigenvalues may be complex.
2) Eigenvectors corresponding to distinct eigenvalues are not necessarily orthogonal.
We solve $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ for each eigenvalue.

Eigenvalue $\lambda_1 = 1$.

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \tag{61}$$

Eigenvalue $\lambda_2 = 1 + i\sqrt{2}$.

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \tag{62}$$

Eigenvalue $\lambda_3 = 1 - i\sqrt{2}$

$$\mathbf{v}_3 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \tag{63}$$

Summary:

$$\lambda_1 = 1, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \tag{64}$$

$$\lambda_2 = 1 + i\sqrt{2}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}, \tag{65}$$

$$\lambda_3 = 1 - i\sqrt{2}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}. \tag{66}$$

Some of the eigen values are complex but the eigen vectors are real.